

Strategic Form Games: Barebones Definitions

An *strategic form game* is a collection $G = (N, \{S_i\}_{i \in N}, \{\succsim_i\}_{i \in N})$ where

1. $N = \{1, \dots, n\}$, a finite set of *players*. For $i \in N$:
2. S_i is player i 's nonempty *strategy set*; and
3. $\succsim_i \subset S \times S$ is player i 's complete and transitive *preference relation* where $S = S_1 \times \dots \times S_n$.

Fix a strategic form game G . For $s \in S$, let s_{-i} be equal to s with the i th component deleted: $s_{-i} = (s_1, \dots, s_{i-1}, s_{i+1}, \dots, s_n)$. As a common abuse of notation I often write $s = (s_i, s_{-i})$. Also let S_{-i} be S with player i 's strategy set removed: $S_{-i} = \prod_{\substack{j=1 \\ j \neq i}}^n S_j$.¹

A strategy $s'_i \in S_i$ is *strictly dominated* for player $i \in N$ in G if there is an $s''_i \in S_i$ such that $(s''_i, s_{-i}) \succ_i (s'_i, s_{-i})$ for every $s_{-i} \in S_{-i}$.² It is *weakly dominated* for player $i \in N$ in G if there is an $s''_i \in S_i$ such that $(s''_i, s_{-i}) \succsim_i (s'_i, s_{-i})$ for every $s_{-i} \in S_{-i}$ and the preference is strict for some $s_{-i} \in S_{-i}$. A strategy $\tilde{s}_i \in S_i$ is *never a best response* for player $i \in N$ in G if for each $s_{-i} \in S_{-i}$, there is a $s_i \in S_i$ with $(s_i, s_{-i}) \succ_i (\tilde{s}_i, s_{-i})$.

Dominant Strategy Equilibrium (DSE)

$$DSE(G) = \{s^* \in S \mid \text{For every } i \in N, (s_i^*, s_{-i}) \succsim_i (s_i, s_{-i}) \text{ for each } (s_i, s_{-i}) \in S_i \times S_{-i}\}.$$

Iterative Deletion of Strictly Dominated Strategies (IDSDS). $IDSDS(G) = \cap_{k=0}^{\infty} \mu^k(S)$, where $\mu^k(S) = \mu(\mu^{k-1}(S))$, $\mu^0 = \mu$, and for $A \subseteq S$, $\mu(A) \subseteq A$ is the set of strategy profiles which remain after deleting all of each player's strictly dominated strategies in $G_A = (N, \{A_i\}_{i \in N}, \{\succsim_{i|A}\}_{i \in N})$, where $A = A_1 \times \dots \times A_n$ and each $\succsim_{i|A}$ is $\succsim_i$ restricted to A .

(Point-) Rationalizable Strategies (R).³ $R(G) = \cap_{k=0}^{\infty} \lambda^k(S)$, where $\lambda^k(S) = \lambda(\lambda^{k-1}(S))$, $\lambda^0 = \lambda$, and for $A \subseteq S$, $\lambda(A) \subseteq A$ is the set of strategy profiles which remain after deleting *all* of *each* player's strategies which are never-a-best-response in $G_A = (N, \{A_i\}_{i \in N}, \{\succsim_{i|A}\}_{i \in N})$.

Nash Equilibrium (NE)

$$NE(G) = \{s^* \in S \mid \text{For every } i \in N, (s_i^*, s_{-i}^*) \succsim_i (s_i, s_{-i}^*) \text{ for each } s_i \in S_i\}.$$

The following facts are easy to show. (For each inclusion there are games G such that the two sets are *not equal*.)

$$DSE(G) \subseteq NE(G) \subseteq R(G) \subseteq IDSDS(G).$$

¹That is $S_{-i} = S_1 \times \dots \times S_{i-1} \times S_{i+1} \times \dots \times S_n$.

²In this case we write that s''_i *strictly dominates* s'_i .

³The qualifier 'point' is from Bernheim, D. "Rationalizable Strategic Behavior," *Econometrica* 52 (1984): 1007-28.